FUNCTIONS QUIZ

BLAKE FARMAN

Lafayette College

Name: Solutions

1. Find the value of b that makes the function

$$f(x) = \begin{cases} \dfrac{x^2-4}{x-2} & \text{if } 2 < x \\ \\ 2x+b & \text{if } x \leq 2 \end{cases}$$

a continuous function.

$$\lim_{x\to 2^+} f(x) = \lim_{x\to 2^+} 2x+b = 4+b = f(2)$$

$$\lim_{x\to 2^-} f(x) = \lim_{x\to 2^-} \frac{x^2-4}{x-2} = \lim_{x\to 2^-} \frac{(x+2)(x-2)}{x-2} = \lim_{x\to 2^-} x+2 = 2+2 = 4$$

To be continuous, we need

$$f(2) = \lim_{x\to 2^+} f(x) = 4+b = \lim_{x\to 2^-} f(x) = 4$$

Therefore $\boxed{b=0.}$